1.3 Fundamental Theorem of Calculus

We first start off with a result based on the average value of a function in the previous section, the Mean Value Theorem for integrals. It states that if we take the average value of a *continuous* function over a closed and bounded interval, then there is a function value at a single point which matches the average value of the function.

Theorem 3 (Mean value theorem for integrals)**.** *Let $f(x)$ be a continuous real-valued function on $[a,b]$. Then there is some $c \in [a,b]$ so that*

$$f(c) = \frac{1}{b-a}\int_a^b f(x)\,dx \qquad \textit{or} \qquad f(c)(b-a) = \int_a^b f(x)\,dx.$$

Picture:

Example 28. Given f_{ave} of $f(x) = 24 - 12x^2$ on $[-2,0]$, find $c \in [-2,0]$ satisfying the mean value theorem.

Given: $f_{\text{ave}} = \dfrac{1}{0-(-2)}\displaystyle\int_{-2}^{0}\left(24-12x^2\right)dx = \dfrac{16}{2}$

Fundamental Theorem of Calculus - Part 1

Picture:

Theorem 4. *For f continuous on $[a, b]$ and $x \in [a, b]$, define the* **accumulation function***:*

$$F(x) = \int_a^x f(t)\, dt.$$

Then $F'(x) = \dfrac{dF}{dx} =$

Note: (**Optional**) For functions f, g, and h, at all well-defined points of continuity, consider the (composition) **accumulation function**:

$$F(x) = \int_{h(x)}^{g(x)} f(t)\, dt.$$

Then $F'(x) = \dfrac{dF}{dx} =$

Fundamental Theorem of Calculus - Part 1

Generalize FTC Part 1: For functions f, g, and h, at all well-defined points of continuity, consider the (composition) **accumulation function**:

$$F(x) = \int_{h(x)}^{g(x)} f(t)\, dt.$$

Then $F'(x) = \dfrac{dF}{dx} =$

Example 29. Differentiation using FTC

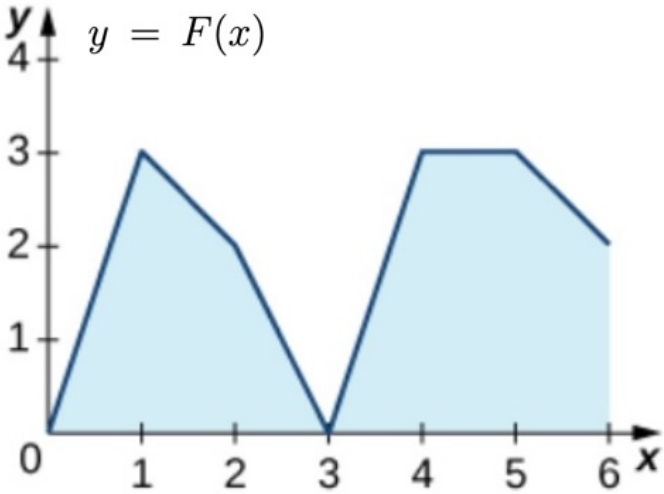
1. Differentiate: $g(x) = \displaystyle\int_{1}^{x} \frac{1}{t^3+1}\, dt.$

2. Differentiate: $h(x) = \displaystyle\int_{\sqrt{x}}^{5} \cos(\pi t)\, dt.$

3. Differentiate: (and simplifty) $F(x) = \displaystyle\int_{x}^{2x} t^3\, dt.$

4. Differentiate: $F(x) = \displaystyle\int_{sin(x)}^{x^3} t^3\, dt.$

Example 30. The graph of the accumulation function $y = F(x) = \int_0^x f(t)\, dt$ is given below. The integrand f is a piecewise constant function.



1. Over which intervals is f positive, negative, or zero?

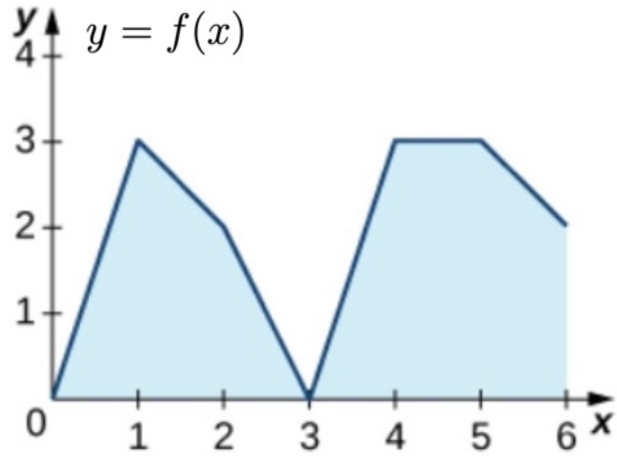
2. What are the maximum and minimum values of f?

3. Evaluate the integral: $\int_1^2 f(x)\, dx =$

4. Evaluate the integral: $\int_3^2 f(x)\, dx =$

5. What is the average vaue of f over the interval $[0, 2]$?

Example 31. Below is the graph of $y = f(x)$. Let $y = F(x) = \int_0^x f(t)\,dt$ be the accumulation function.



1. Represent $F(1)$ as an integral and evaluate it.

2. Over which intervals is f positive, negative, or zero?

3. What are the maximum and minimum values of f?

4. What are the maximum and minimum values of F?

5. Evaluate the integral: $\int_1^2 f(x)\,dx =$

6. Determine $\dfrac{dF}{dx}(4)$

7. Determine $F''(4)$

8. What is the average vaue of f over the interval $[0, 2]$?

FTC - Part 2

Theorem 5 (FTC Part 2). *let $y = f(x)$ continuous on $[a, b]$ and $F(x)$ be any antiderivative of $f(x)$, then*

$$\int_a^b f(t)\,dt = \quad F(x)\Big|_a^b \quad = \quad F(b) - F(a)$$

Example 32. Basic antiderivatives:

1. $\displaystyle\int x^n\,dx = \frac{x^{n+1}}{n+1} + C$
2. $\displaystyle\int \cos(x)\,dx = \sin(x) + C$
3. $\displaystyle\int \sin(x)\,dx = -\cos(x) + C$
4. $\displaystyle\int \sec^2(x)\,dx = \tan(x) + C$
5. $\displaystyle\int e^x\,dx = e^x + C$
6. $\displaystyle\int \frac{1}{x}\,dx = \ln(\,|x|\,) + C$

Example 33. Evaluate the **definite integrals**:

1. $\displaystyle\int_0^{\pi} \Big(t^2 + 4 - \sin(t)\Big)\,dt =$

2. $\displaystyle\int_1^2 \left(\frac{x^2 - 1}{\sqrt{x}} + (x+1)(x-1)\right)dx =$

Example 34. Evaluate the following **definite integrals**:

1. $\displaystyle\int_0^1 \left(t^2 + \sqrt{t}\right) dt =$

2. $\displaystyle\int_{-1}^1 \left(r + 1\right)^2 dr =$

3. $\displaystyle\int_1^2 \frac{y^5 - 2y}{y^3}\, dy =$

4. $\displaystyle\int_0^2 \left|2x - 2\right| dx =$

4.10 Antiderivatives and Initial-Value Problems

Reading: Content from Section 4.10 is found in OpenStax Volume 1.

Link: https://openstax.org/books/calculus-volume-1/pages/4-10-antiderivatives

Antiderivatives

How do you "undo" a derivative?

Definition 4. A function F is an antiderivative of f on an interval I provided

$$F'(x) = f(x) \qquad \text{for all } x \text{ in } I.$$

Example 35. Is $H(x) = x^2 + 7x + 2$ and antiderivative of $f(x) = 2x + 7$?

What about $G(x) = 3x^2 + 7x - 20$?

Example 36. Identify the family of antiderivatives for $f(x) = 2x + 7$.

Example 37. Evaluate the following **indefinite integrals**:

1. $\displaystyle\int \frac{2}{\pi}\sin(x)\,dx =$

2. $\displaystyle\int \left(e^x - \sec^2(x)\right)\,dx =$

3. $\displaystyle\int \left(\frac{1}{x} + \frac{1}{x^2}\right)\,dx =$

4. (Algebra, poly long division, and integral table)

$$\int \frac{x^2}{1+x^2}\,dx =$$

Example 38. Solve the following **initial value problems.**

1. A particle moves forward and backward in a straight line and has velocity given by $v(t) = -2t^2 - 3t + 6$, in meters (m) per second (s). After 1 second, the particle is 6 meters forward, i.e. $s(1) = 6\,m$. Find its position function $s(t)$.

2. A car is being driven along a highway. At the moment the car brakes are applied, the car was traveling at a rate of 12 m/s. The car decelerates at a constant rate of 3 m/s^2 until it stops. From the instant the brakes were applied, how far does the car travel before it stops?

Example 39. Solve various initial value problems from below:

<table>
<tr><td>1. $\frac{dy}{dx} = 5x^4 - \sec^2 x$</td><td>2. $\frac{dy}{dx} = \sin x - e^{-x} + 8x^3$</td><td>3. $\frac{dy}{dx} = \frac{1}{x} - \frac{1}{x^2}$, $(x > 0)$</td><td>4. $\frac{dy}{dt} = 3t^2 \cos\left(t^3\right)$</td></tr>
<tr><td>5. $\frac{dy}{dx} = 3\sin x$;
$y = 2$ when $x = 0$</td><td>6. $\frac{dy}{dx} = 2e^x - \cos x$;
$y = 3$ when $x = 0$</td><td>7. $f'(x) = 7x^6 - 3x^2 + 5$;
$f(1) = 1$</td><td>8.
$y' = 10x^9 + 5x^4 - 2x + 4$;
$y = 6$ when $x = 1$</td></tr>
<tr><td>9. $f'(x) = -\frac{1}{x^2} - \frac{3}{x^4} + 12$
$f(1) = 3$</td><td>10. $\frac{dy}{dx} = 5\sec^2 x - \frac{3}{2}\sqrt{x}$;
$y\big\vert_{x=0} = 7$</td><td>11. $\frac{dx}{dt} = \frac{1}{t} - \frac{1}{t^2} + 6$;
$x = 0$ when $t = 1$</td><td>12.
$\frac{dv}{dt} = 4\sec t \tan t + e^t + 6t$;
$v = 5$ when $t = 0$</td></tr>
<tr><td colspan="2">13. Find the specific solution to the following second-order initial value problem by first finding $\frac{dy}{dx}$ and then finding y:

$\frac{d^2y}{dx^2} = 24x^2 - 10$. When $x = 1$, $\frac{dy}{dx} = 3$ and $y = 5$.</td><td colspan="2">14. Find the specific solution to the following second-order initial value problem by first finding $f'(x)$ and then finding $f(x)$:

$f''(x) = \cos x - \sin x$. $f'(0) = 2$ and $f(0) = 0$</td></tr>
</table>

1.4 Net Change, Velocity, Change

Definition 5. Net and total change formulas:

1. Displacement (net change in distance):

$$s(b) - s(a) = \int_a^b v(t)\, dt$$

2. Distance traveled (total distance along the path):

$$\int_a^b |v(t)|\, dt$$

3. Position from velocity:

$$s(t) = s(0) + \int_0^t v(x)\, dx$$

4. Net change of Q from $t = a$ to $t = b$:

$$Q(b) - Q(a) = \int_a^b Q'(t)\, dt$$

Example 40. Cells start with an initial population 100 and increase at a rate $N'(t) = 90e^{-0.1t}$. Find a formula for number of cells $N(t)$.

Example 41. Given velocity $v(t) = 3t^2 - 12$ (in meters per second) for an object in motion from time $t = 0$ to time $t = 3$, find the (net) displacement.

Example 42. Given velocity $v(t) = 3t^2 - 12$ (in meters per second) for an object in motion from time $t = 0$ to time $t = 3$, find the (total) distance travelled.

Example 43. A block hangs on a spring at the origin $s = 0$. At $t = 0$ it is pulled down $\frac{1}{4}\,m$ to its initial position. Its velocity is $v(t) = \frac{1}{4}\sin(t)$, for $t \geq 0$. Assume up is positive, set up the formula needed (using integrals) to determine the position $s(t)$. **Do NOT** evaluate the integral.

Example 44. A skydiver jumps from a plane. Almost immediately, she hits terminal velocity at $t = 0$ of $80\,m/s$. At 19 seconds the parachute opens and the velocity continues as shown below. Approximate the height of the jump.

$$v(t) = \begin{cases} 80 \;, & \text{if } 0 \leq t \leq 19 \\ 783 - 37t \;, & \text{if } 19 \leq t \leq 21 \\ 6 \;, & \text{if } 21 \leq t \leq 40 \end{cases}$$

Symmetry properties

Definition 6. A couple symmetries:

- Odd function: $f(-x) = -f(x)$, so

  $$-- > \quad \int_{-a}^{a} f(x)\,dx = 0$$

- Even function: $f(-x) = f(x)$, so

  $$-- > \quad \int_{-a}^{a} f(x)\,dx = 2\int_{0}^{a} f(x)\,dx$$

Example 45. Discuss "simplifying" the integrals below using symmetry:

1. $\displaystyle\int_{-2}^{2} x^4 - 3x^3\,dx$

2. $\displaystyle\int_{-\pi/2}^{\pi/2} \cos(x) - 4\sin^3(x)\,dx$

Example 46. Suppose f is an even function, $\displaystyle\int_{-2}^{2} f(x)\,dx = \sqrt{2}+4$, and $\displaystyle\int_{-2}^{1} f(x)\,dx = 2$,

1. Evaluate $\displaystyle\int_{-2}^{0} f(x)\,dx =$

2. Evaluate $\displaystyle\int_{-2}^{-1} f(x)\,dx =$

1.5 Integration by substitution

or the Chain Rule for Integrals

Consider the following integrals:

$$\int \frac{e^x}{e^x+1}\,dx \qquad \text{and} \qquad \int x^2\cos(x)\,dx$$

Note: This technique helps us evaluate more integrals, but there are some we still cannot.

Recall the chain rule: For $y = f(u(x))$, we have $y' = f'(u(x)) \cdot u'(x)$.

Theorem 6.

$$\int f'(u(x)) \cdot u'(x)\,dx = f(u(x)) + C$$

Example 47. Evaluate by matching components with those in the theorem above:

1. $\displaystyle\int \cos(x^2)(2x)\,dx =$

2. $\displaystyle\int \sin(4x^3+x)(12x^2+1)\,dx =$

Terminology: This technique is ofter referred to as: u-substitution.

Recall: By FTC,

$$\int f'(u)\,du = f(u) + C.$$

Definition 7. du is the differential given by

$$\frac{du}{dx} = u'(x) \quad \Leftrightarrow \quad du = u'(x)dx.$$

Example 48. Find $\frac{du}{dx}$. Then solve for du.

1. $u = x^2$

2. $u = \sin(x)$

Example 49. Evaluate these using the above techniques.

1. $\int \cos(x^2)(2x)\,dx =$

2. $\int \sin^3(x)\cos(x)\,dx =$

Example 50. Discuss strategies for finding $u = u(x)$:

1. $\displaystyle\int \cos(x)e^{\sin(x)}\,dx$

2. $\displaystyle\int \sin^5(x)\cos(x)\,dx$

3. $\displaystyle\int \frac{\cos(x)}{\sin(x)}\,dx$

Theorem 7.

$$\int \cot(x)\,dx = -\ln|\csc(x)| + C$$

1.5 Integration by substitution, continued

Definite Integrals with Substitution

Theorem 8. *Suppose u is differentiable on (a,b), continuous on $[a,b]$, and f is continuous on the range of u. Then*

1. *Leaving the bounds of integration:*

$$\int_a^b f(u(x))\cdot u'(x)\,dx = \int_{x=a}^{x=b} f(u)\,du = \quad F(u)\Big|_{x=a}^{x=b} = \quad F(u(x))\Big|_a^b$$
$$= F(u(b)) - F(u(a))$$

2. *Changing the bounds:*

$$\int_a^b f(u(x))\cdot u'(x)\,dx = \int_{u(a)}^{u(b)} f(u)\,du = \quad F(u)\Big|_{(u(a)}^{u(b)} = F(u(b)) - F(u(a))$$

Example 51. Integrate using the associated method:

Method 1. $\displaystyle\int_0^\pi \cos(x^2)(2x)\,dx$

Method 2. $\displaystyle\int_0^\pi \cos(x^2)(2x)\,dx$

Example 52. If possible, identify $u(x)$ and solve.

1. $\displaystyle\int_0^1 \frac{x}{3x^2+1}\,dx$

2. $\displaystyle\int \sin^2(x)\cos(x)\,dx$

3. $\displaystyle\int \cos(x)\tan(x)\,dx$

4. $\displaystyle\int x^6 e^{5x^7-2}\,dx$

5. (Back sub) $\displaystyle\int 5x\sqrt{x+1}\,dx$

6. (Back sub) $\displaystyle\int 15x^9\sqrt{x^5+1}\,dx$

7. $\int \frac{\sin(\sqrt{x})}{\sqrt{x}}\,dx$

Example 53. Evaluate the indefinite integral given below.

$$\int -x^2 \sin^5(5x^3)\cos(5x^3)\,dx$$

Use substitution to evaluate the definite integral given below.

$$\int_0^{\sqrt{\frac{\pi}{8}}} 5x \tan^4(2x^2)\sec^2(2x^2)\,dx$$

Example 54. 1. Evaluate the indefinite integral given below.

$$\int \frac{4x}{(4x^2-7)^6}\,dx$$

2. Evaluate the indefinition integral given below.

$$\int -\frac{2x^2}{(4-5x^3)^7}\,dx$$

Section 1.6: Integrals Involving Exponentials and Logarithms

Theorem 9 (Rules involving exponentials)**.** *Two general rules for exponentials:*

1. $\displaystyle\int e^x \, dx = e^x + C$
2. $\displaystyle\int a^x \, dx = \frac{a^x}{\ln(a)} + C$

Example 55. Evaluate

1. $\displaystyle\int e^{-x} \, dx$

2. $\displaystyle\int e^x \sqrt{1 + e^x} \, dx$

3. $\displaystyle\int 7x^2 e^{x^3} \, dx$

Section 1.6: Integrals Involving Exponentials and Logarithms, continued

Example 56. (Optional) The rate of growth of bacteria in a petri dish is $q(t) = 3^t$, with t in hours and $q(t)$ in thousands of bacteria per hour. This culture starts with 10,000 bacteria. Find $Q(t)$, a function that give the number of bacteria in the petri dish. How many bacteria are there after 2 hours?

Theorem 10 (Rules involving logarithms)**.** *1.* $\displaystyle\int \frac{1}{x}\,dx = \int x^{-1}\,dx =$

2. $\displaystyle\int \frac{f'(x)}{f(x)} dx =$

3. $\displaystyle\int \ln(x)\,dx = x(\ln(x) - 1) + C$

4. $\displaystyle\int \log_a(x) = \frac{x}{\ln(a)}(\ln(x) - 1) + C$

Example 57. Evaluate by finding antiderivatives.

1. $\displaystyle\int \frac{2x^3 + 3x}{x^4 + 3x^2}\,dx$

2. $\displaystyle\int_0^{\frac{\pi}{2}} \frac{\sin(x)}{1 + \cos(x)}\,dx$

Example 58. Evaluate the indefinite integral given below.

$$\int 5\tan(2x)\ln(\cos(2x))\,dx$$

Example 59. Evaluate the indefinite integral given below.

$$\int \frac{-3e^{-6/x^5}}{x^6}\,dx$$

Section 1.7: Inverse Trig Functions from integral table

Section 1.7 NOT on Mini-Exam 2.

1. $\displaystyle\int \frac{f'(x)}{\sqrt{1-f(x)^2}}\,dx = \sin^{-1}(f(x)) + C$

2. $\displaystyle\int \frac{f'(x)}{1+f(x)^2}\,dx = \tan^{-1}(f(x)) + C$

3. $\displaystyle\int \frac{f'(x)}{|f(x)|\sqrt{f(x)^2-1}}\,dx = \sec^{-1}(f(x)) + C$

Example 60. Apply the above integrals after identifying $f(x)$:

1. $\displaystyle\int \frac{1}{\sqrt{1-x^2}}\,dx =$

2. $\displaystyle\int \frac{1}{|x|\sqrt{x^2-1}}\,dx =$

3. $\displaystyle\int \frac{x}{1+x^4}\,dx =$

4. $\displaystyle\int \frac{3}{\sqrt{1-(3x+1)^2}}\,dx =$